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- Introduction to Module 9's Study
Video • 2 min
- Why Data Preparation Is Not Optional in
Machine Learning
Reading • 5 min

- ✓ Basic Data Cleaning Review
Video • 38 sec
- ✓ Materials on Basic Data Cleaning
Reading • 10 min
- ✓ Cleaning Data Is a Decision, Not a Checklist
Reading • 5 min

- ✔ **Distributions, Outliers and Their Removal:**
Gaussian Distribution, Skewness and Outliers
Video • 6 min
- ✔ **Distributions, Outliers and Their Removal: The**
Z-Score and IQR Method for Outliers Removal
Video • 5 min
- ✔ **Materials on Distributions, Outliers and Their**
Removal
Reading • 10 min
- **Outliers, Assumptions, and When "Cleaning"**
Becomes Damage
Reading • 5 min

- ✔ Data Transform: Normalization, Standardization, Power Transform
Video • 7 min
- ✔ Materials on Data Transform
Reading • 10 min
- When Scaling Helps — and When It Quietly Breaks Your Model
Reading • 5 min

- ✓ Construct Training and Test Sets for Model Evaluation: Methods, Implementation, and Stratified Sampling
Video • 5 min
- ✓ Construct Training and Test Sets for Model Evaluation: Cross-Validation
Video • 6 min
- ✓ Materials on Training/Test Sets Generation
Reading • 10 min
- ✓ Evaluation Starts with the Split: When Performance Numbers Lie
Reading • 5 min

- ✔ Data Preparation Overview
Video • 3 min
- ✔ Materials on Data Preparation
Reading • 10 min
- From Raw Data to Reliable Learning: What Actually Matters?
Reading • 5 min
- Data Preparation and Evaluation Coach
Dialogue • 5 min

- Module 9 Quiz
Practice Assignment • 30 min

- Module 9 Assignment 1
Graded App Item • 30 min
- Module 9 Assignment 2
Graded App Item • 30 min
- Module 9 Assignment 3
Graded App Item • 30 min
- XLS file for Module 9 Assignment
Reading • 1 min
- Module 9 Recap
Reading • 5 min

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- Introduction to Module 10's Study Video • 1 min
- Why Support Vector Machines? Reading • 5 min

- Support Vector Machine Fundamentals:
Concets

From Raw Data to Reliable Learning: What Actually Matters?



- Reflect on why data preparation steps such as cleaning, outlier handling, transformation, scaling, and train/test splitting should be treated as parts of a connected pipeline rather than isolated tasks.
- Consider what kinds of problems could occur if these preprocessing steps are applied in the wrong order.

- outlier detection,
- data transformation, and
- train/test splitting.
- Consider the consequences that could arise if any of these assumptions are violated in practice.

- Reflect on whether aggressive data preparation could unintentionally reduce model performance or harm generalization.
- Consider an example where making the dataset “cleaner” could actually remove important information or reduce the model’s ability to handle real-world variability.

- After completing this module, reflect on the checks or preprocessing steps you would now perform before applying any machine learning algorithm to a new dataset.
- Consider which data preparation step you believe is most critical for ensuring reliable and trustworthy model evaluation, and explain why.

- Reflect on how poor data preparation may affect more advanced machine learning models such as support vector machines or neural networks.
- Consider why the risks associated with poor-quality data may increase as models become more flexible, expressive, and capable of fitting complex patterns.

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